

Waterflood injection recommendation

streak_5x4 sample · asset SAMPLE · unit system field

HIGH confidence objective: maximum NPV · posture: balanced · horizon: 24 months

Approved by originator — the person who ran this recommendation also approved it (separation policy: log).

Sector S1 — summary

confidence	HIGH blind R ² 1.00; parameter spread 0%; no gate warnings
method	CRMP (producer) (24 parameters)
blind-test R ² / MAPE	1.000 / 0.0 %
gain vs hold-current	+1.1 %
gain vs equal split	+1.3 %
plan cumulative oil P10 / P50 / P90	24,393 / 24,393 / 24,393 bbl
hold-current P10 / P50 / P90	23,925 / 23,925 / 23,925 bbl
posture	balanced

Change list — this week

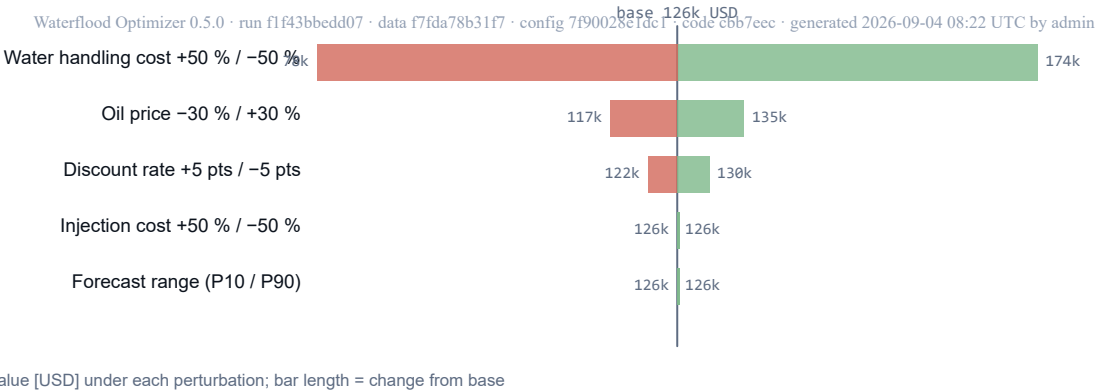
injector	from [bbl/d]	to [bbl/d]	step this week	target date	setting hint	expected oil [bbl]	revert if
I-3	1,841	0	-276	2023-01-15	target 0; set choke / pump frequency to reach it, field to confirm the setting	124	water cut of any connected producer rises by more than 5%
I-4	1,646	0	-247	2023-01-08	target 0; set choke / pump frequency to reach it, field to confirm the setting	111	water cut of any connected producer rises by more than 5%
I-1	2,310	3,612	346	2019-12-29	target 3612; set choke / pump frequency to reach it, field to confirm the setting	87	water cut of any connected producer rises by more than 5%
I-5	1,450	2,695	217	2020-01-05	target 2695; set choke / pump frequency to reach it, field to confirm the setting	84	water cut of any connected producer rises by more than 5%
I-2	1,782	2,722	267	2019-12-29	target 2722; set choke / pump frequency to reach it, field to confirm the setting	63	water cut of any connected producer rises by more than 5%

Cumulative oil over the horizon: plan vs hold-current (P10 – P50 – P90)

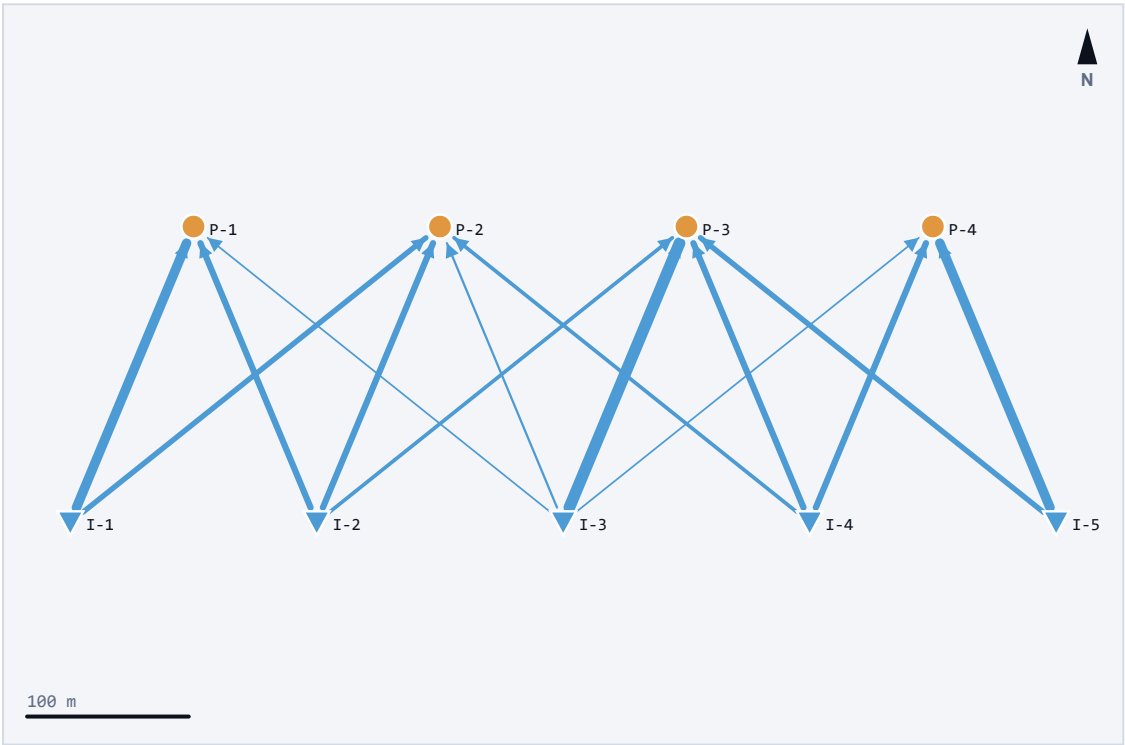


[bbl]; bars span P10–P90, the tick is P50

Sensitivity of the gain over hold-current



Connectivity



▼ injector ● producer ● mixed — f_{ij} (width $\propto f$, opacity $\propto$ confidence)

Connectivity f_{ij} (rows: injectors, columns: producers)

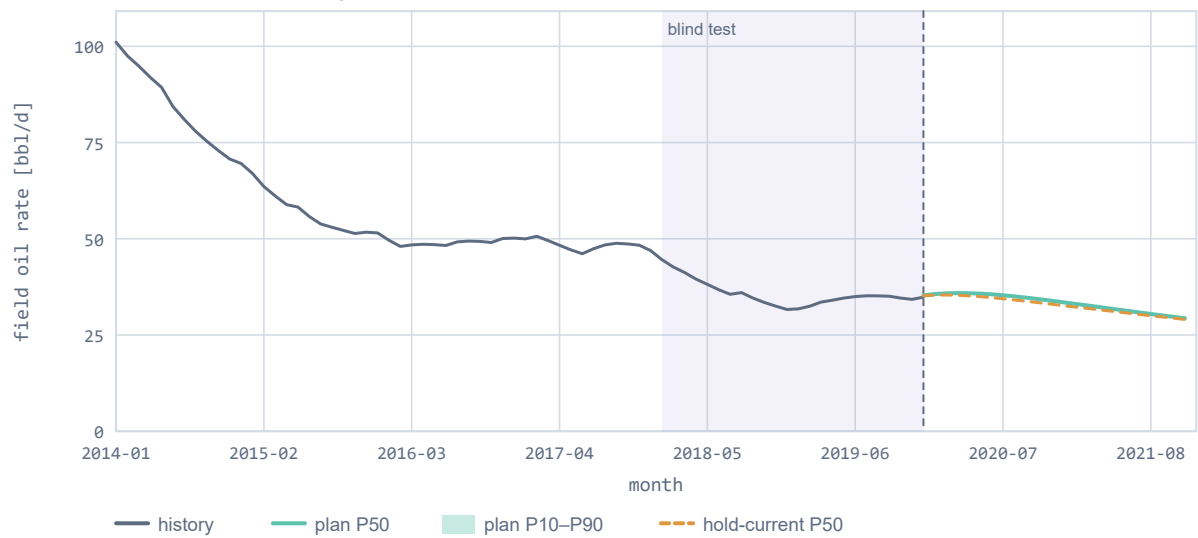
	P-1	P-2	P-3	P-4
I-1	0.60	0.30		
I-2	0.36	0.36	0.18	
I-3	0.04	0.07	0.75	0.04
I-4		0.18	0.36	0.36
I-5			0.30	0.60
τ [d]	150	110	95	180

$\sum_j f_{ij}$ per injector: I-1 0.90, I-2 0.90, I-3 0.90, I-4 0.90, I-5 0.90

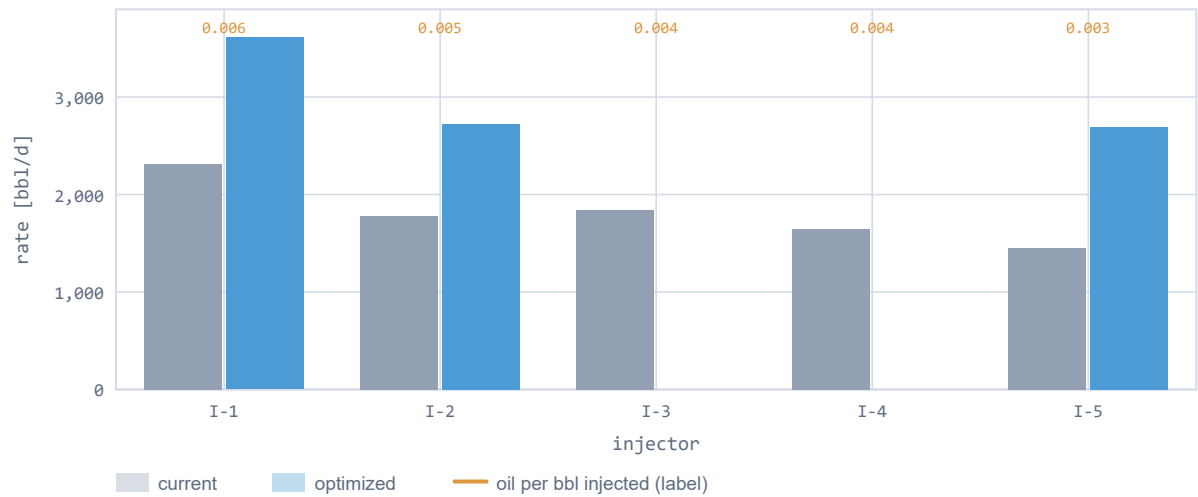
Forecast and injector efficiency

Waterflood Optimizer 0.5.0 · run f1643bbed107 · data f7fda78b31f7 · config 7f90028e1dc1 · code cbb7ecc · generated 2026-09-04 08:22 UTC by admin

Field oil rate: history and forecast



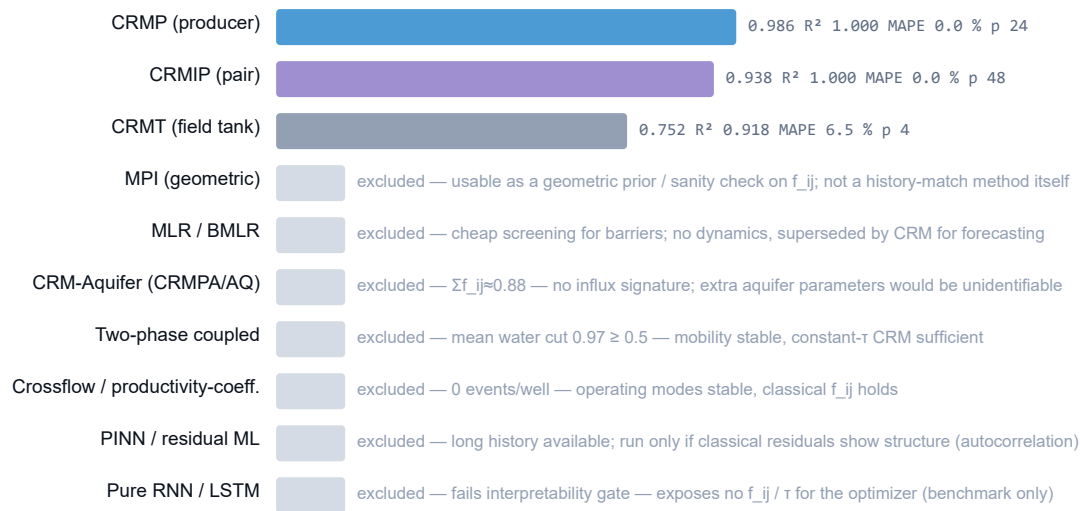
Injector efficiency (oil per bbl injected) — current vs optimized rate



Which method won

Waterflood Optimizer 0.5.0 - pos: f1643bbed107 - data: 76b78b3157 - conf: 730028 - id: 1 - code: bbb7c8c - generated 2026-03-04 08:22 UTC by admin

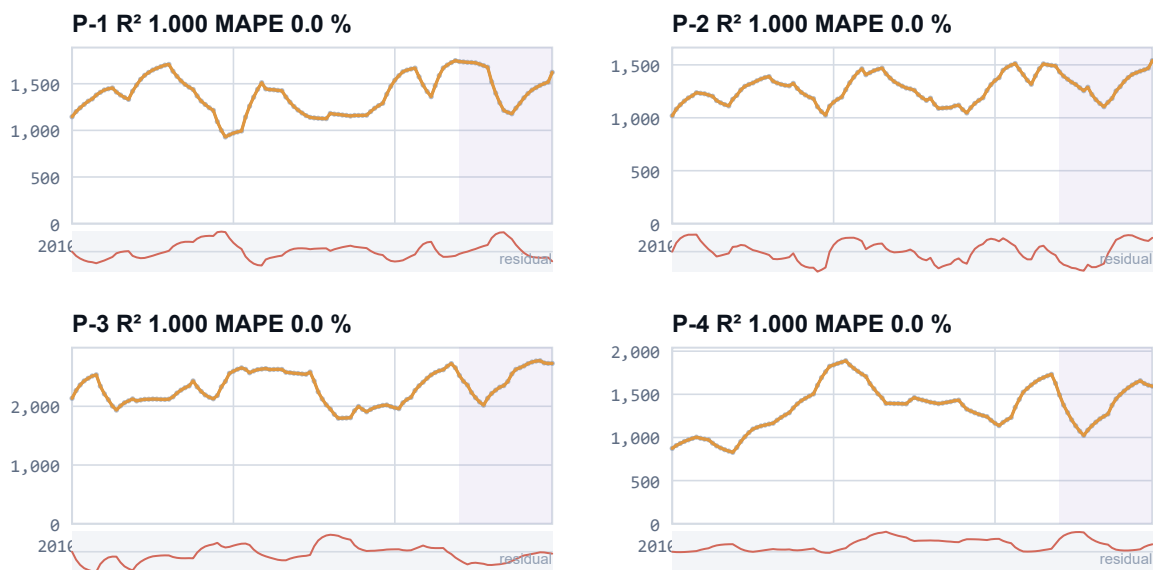
Which method won — composite score (blind R², MAPE, AICC, stability, plausibility, runtime)



gate	result	gate	result
history_min	pass	od	pass
points_per_param	pass	crmip_data	pass
inj_cv	pass	tau_dt	pass
influx	fail / n.a.	event_heavy	fail / n.a.
mature	pass	long_history	pass
has_xy	pass	has_bhp	fail / n.a.
has_relperm	fail / n.a.		

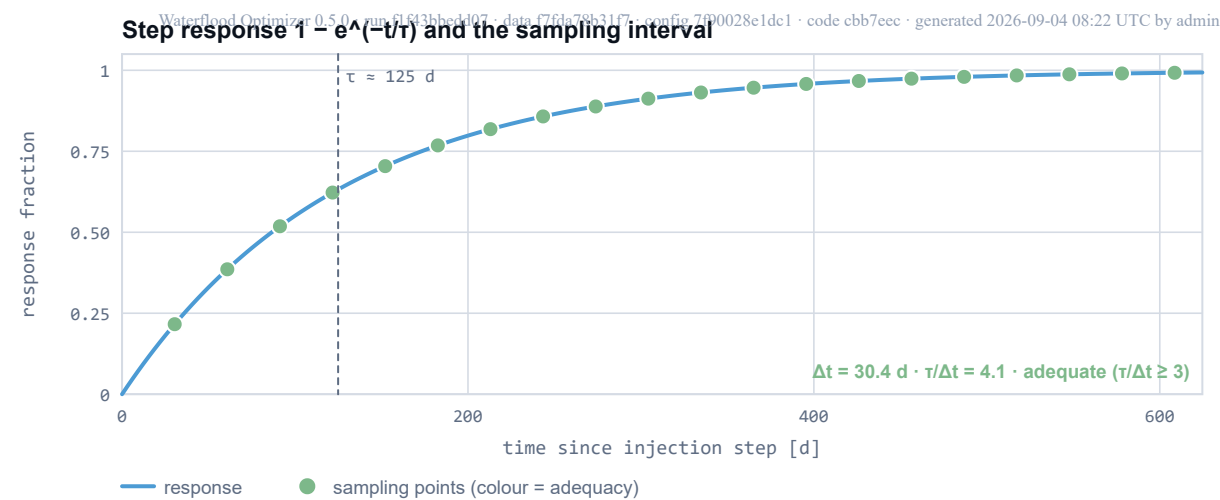
Excluded methods: aquifer — $\Sigma f_{ij} \approx 0.88$ — no influx signature; extra aquifer parameters would be unidentifiable; twophase — mean water cut $0.97 \geq 0.5$ — mobility stable, constant- τ CRM sufficient; crossflow — 0 events/well — operating modes stable, classical f_{ij} holds; rnn — fails interpretability gate — exposes no f_{ij} / τ for the optimizer (benchmark only)

History match per well



liquid rate [bbl/d] · raw (points), cleaned (grey), model (orange); blind window shaded

Sampling and pressure



$\Delta t = 30 \text{ d} \cdot \tau \text{ estimate} = 125 \text{ d} \cdot \tau/\Delta t = 4.1 \cdot O_d = 20.0 \cdot \text{monthly allocated data} \cdot \text{field pressure source: none}$

Data quality

Traffic light: green · steps 120 · outliers removed 0 · simultaneous P+I steps 0 · unmatched IDs 0 · warnings 0 · errors 0

well	type	coverage %	pressure	conversions
I-1	INJ	100		
I-2	INJ	100		
I-3	INJ	100		
I-4	INJ	100		
I-5	INJ	100		
P-1	PROD	100	none	
P-2	PROD	100	none	
P-3	PROD	100	none	
P-4	PROD	100	none	

Conditions (plain language)

severity	what it means	suggested action	scope
info	Reservoir-volume conversion uses constant formation-volume factors ($B_o = 1.0$, $B_w = 1.0$).	Load a PVT table if formation volume factors change with pressure.	field
info	No flowing pressure data; lift or choke changes must be in the events table.	Load ESP/gauge pressure or an events log if lift settings changed during the history.	field
info	The MPI (geometric) model would suit this data but is not available in this release.	None needed; the next best available model was used.	field
info	The MLR / BMLR model would suit this data but is not available in this release.	None needed; the next best available model was used.	field
info	The PINN / residual ML model would suit this data but is not available in this release.	None needed; the next best available model was used.	field
info	The model misses some slow trends for P-1; forecasts may drift.	None needed for screening; a residual correction is considered when history exceeds 120 months.	well:P-1
info	The model misses some slow trends for P-2; forecasts may drift.	None needed for screening; a residual correction is considered when history exceeds 120 months.	well:P-2
info	The model misses some slow trends for P-3; forecasts may drift.	None needed for screening; a residual correction is considered when history exceeds 120 months.	well:P-3

severity	what it means	suggested action	scope
info	The model misses some slow trends for P-4; forecasts may drift.	None needed for screening; a residual correction is considered when history exceeds 120 months.	well:P-4

Workflow

recommendation	422b38fb649a
state	APPROVED
originator	admin
approved by originator	YES — same person ran and approved
override reason	—
advanced overrides	none
snapshot digest	6a0b517bad5e5f2fa974da02b1b8e8ce338281d5c7e683826a350295d6d61cc6
implemented at	—

when	action	actor	acting role	from → to	note
2026-09-04 08:22	review	admin	reviewer	DRAFT → REVIEWED	sample review
2026-09-04 08:22	approve	admin	approver	REVIEWED → APPROVED	sample approval

injector	recommended [bbl/d]	implemented [bbl/d]
I-1	3,612	—
I-2	2,722	—
I-3	0	—
I-4	0	—
I-5	2,695	—

Reproducibility

run id	f1f43bbedd07
data snapshot hash	f7fda78b31f7813fabf5d9e653ed8508b3f642bb382c1ebbbf405b4a1ddc592e
config hash	7f90028e1dc16a82f17887a8f06f66883030895f1fc53860299672b9c6576ee0
code version	cbb7eec
seed	232413243
engine runtime	22.3 s
windows	2010-01-01 → 2019-12-01 (full history)
application	Waterflood Optimizer 0.5.0 — CRM-based waterflood evaluation and optimization (architecture v2.1)